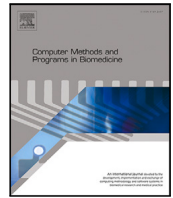




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Reference Vector-guided Evolutionary Algorithm for cluster analysis of single-cell transcriptomes

Fernando M. Rodríguez-Bejarano¹, Miguel A. Vega-Rodríguez¹*, Sergio Santander-Jiménez¹*Escuela Politécnica, Universidad de Extremadura¹, Campus Universitario s/n, 10003 Cáceres, Spain*

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ABSTRACT

Background and Objective: Single-cell RNA-sequencing (scRNA-seq) has revolutionized transcriptomic studies by providing detailed insights into gene expression profiles at the single-cell level. This technology allows researchers to capture expression patterns of thousands of genes across hundreds or thousands of individual cells. Clustering is a crucial step in the analysis of scRNA-seq data, since it enables the identification of distinct cell populations based on their transcriptomic profiles and serves as a foundation for downstream analysis. Given that clustering scRNA-seq data is a challenging task that involves different conflicting objectives, our goal is to tackle it from a multi-objective optimization perspective.**Methods:** This study proposes a Reference Vector-guided Evolutionary Algorithm for Cluster Analysis of Single-cell Transcriptomes (RVEA-CAST) to address the clustering task as a multi-objective optimization problem. Our approach considers three objectives to optimize: clustering deviation, clustering compactness, and the Davies–Bouldin index. The algorithmic design of RVEA-CAST incorporates three problem-aware mutation operators specifically designed to improve each objective, which are orchestrated under a multi-objective search engine based on the use of reference vectors.**Results:** RVEA-CAST is evaluated on ten real scRNA-seq datasets using standard clustering evaluation metrics, such as Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI). The attained results reveal the improved performance and robustness of the proposed approach compared to other previously proposed methods. Specifically, statistically significant improvements of up to 66.7% and 261.5% were achieved for NMI and ARI, respectively. Furthermore, the analysis of differentially expressed genes in the predicted and real clusters showcased greater agreement of our solutions with actual cell populations, underscoring the biological relevance of our approach.**Conclusions:** The results highlight that RVEA-CAST is an effective and versatile approach for clustering scRNA-seq data, outperforming existing methods across diverse biological scenarios in both widely used clustering evaluation metrics and biological relevance.

1. Introduction

In the last two decades, bulk RNA sequencing has been the primary tool for transcriptomic studies, providing averaged gene expression data across cell populations within a biological sample. However, the emergence of single-cell RNA sequencing (scRNA-seq) technologies has promoted new advances in the field by enabling transcriptome profiling at the single-cell level. This approach has revolutionized biomedical research by unraveling cellular heterogeneity and revealing distinct cell types or states in complex biological samples [1]. Nevertheless, due to the low amount of transcripts in a single cell in comparison with bulk RNA-seq, this technique has to face problems related to a high level of

noise and dropouts, in addition to other downsides such as the presence of doublets [2].

Clustering single-cell transcriptomes is a crucial step in scRNA-seq analysis, facilitating the identification of novel cell populations and differentially expressed genes within specific cell clusters. This process involves grouping cells based on their transcriptomic profiles, where cells belonging to a particular population share similar profiles and differentiate themselves from other populations [3]. However, the clustering step must face specific issues, including computational costs, especially with datasets composed of thousands of cells, as well as challenges in visualization and interpretation. Selecting an appropriate

* Corresponding author.

E-mail addresses: fernandorb@unex.es (F.M. Rodríguez-Bejarano), mavega@unex.es (M.A. Vega-Rodríguez), sesaji@unex.es (S. Santander-Jiménez).¹ <https://ror.org/0174shg90>

number of clusters significantly impacts the outcome, making it a difficult task. Moreover, validating clustering solutions becomes complex in the absence of reliable ground truth and benchmark datasets [4].

Clustering scRNA-seq data represents a multi-objective problem where multiple competing objectives need to be optimized simultaneously. The main goal is to group cells into distinct clusters based on their transcriptomic profiles, satisfying multiple quality goals. A first objective that can be identified in this context is to minimize within-cluster deviation, which can benefit from increasing the number of clusters to obtain smaller, more compact clusters. On the other side, reducing the number of clusters can be advantageous to boost other objectives, such as the Davies–Bouldin index (DBI), an optimality criterion that takes into account distances between clusters. Thus, formulating this problem as a multi-objective one, the trade-offs between competing objectives can be explored in order to generate a set of clustering solutions that balance them.

In this work, we introduce a novel evolutionary approach to conduct clustering tasks in scRNA-seq data through a multi-objective optimization perspective: the Reference Vector-guided Evolutionary Algorithm for Cluster Analysis of Single-cell Transcriptomes (RVEA-CAST). Our algorithm defines multiple problem-aware mutation operators which have been specifically designed to improve each objective of the problem, thereby enhancing the overall clustering performance. Our method has been extensively tested across different biological contexts using ten real scRNA-seq datasets with a different range of cell counts, genes, and cell types. The results obtained by RVEA-CAST have been compared with various state-of-the-art methods to evaluate the relevance of the proposed approach in terms of clustering accuracy. For this purpose, standard evaluation metrics commonly employed in the clustering field have been adopted to perform the comparison of clustering solutions. To provide a biological insight into the results, differential gene expression analysis has also been performed between the clusters predicted by each method and the real clusters. The differentially expressed genes have been compared employing binary classification metrics to assess the relevance of RVEA-CAST. Therefore, the main contributions of this work can be summarized as follows:

- A novel multi-objective optimization algorithm for cluster analysis of single-cell transcriptomes is proposed.
- Three problem-aware mutation operators are defined to specifically improve each objective of the problem.
- The performance of our method has been thoroughly evaluated with ten real scRNA-seq datasets with varying biological scenarios.
- The results obtained by RVEA-CAST have been compared with various state-of-the-art methods previously proposed by other authors.
- Rigorous evaluation metrics widely used in the field highlight that RVEA-CAST consistently outperforms other state-of-the-art methods.
- The biological analysis of the clustering solutions denotes a high agreement of our method with the real clusters.

This paper is organized as follows. Section 2 gathers the previously proposed multi-objective evolutionary algorithms in the field of cluster analysis employing scRNA-seq data. The definition of the objectives that compose the problem of cluster analysis of single-cell transcriptomes is detailed in Section 3. Section 4 describes the algorithmic strategies and techniques included in RVEA-CAST and its mutation operators. Section 5 provides an overview of the employed datasets, experimental settings, and evaluation metrics, as well as a comparison of the results obtained by RVEA-CAST with other methods, an analysis of the importance of the intelligent mutation operators, an assessment of RVEA-CAST's scalability, and a biological analysis. Finally, Section 6 outlines the conclusions and significance of our method for clustering analysis of single-cell transcriptomes and proposes future work.

2. Related work

Clustering is a popular unsupervised machine learning technique that groups distinct sets of instances based on their similarity. In bioinformatics, where the analysis of large-scale biological data is essential, clustering plays a fundamental role. Specifically, it is applied to identify protein complexes within protein–protein interaction networks [5], to cluster genes for the detection of genomic islands [6,7], and to group sequences into operational taxonomic units for microbiome analysis [8]. Interestingly, clustering has also been used to classify nanopore current signals, aiding in the identification of DNA methylation sites [9]. Consequently, clustering has emerged as a crucial tool in bioinformatics, facilitating the analysis of high-dimensional datasets, including those generated by scRNA-seq technologies.

In particular, clustering analysis is essential in scRNA-seq for discovering the cellular composition and functional diversity of biological samples. This observation has led to the widespread use of this approach to identify cell types based on their transcriptomic profiles, facilitating the creation of different atlas projects, such as the Human Cell Atlas [10,11]. These atlases aim to identify and describe every cell type, providing a comprehensive reference map that can be leveraged to discover new cell types and understand cellular interactions. This approach also extends to the field of personalized medicine, where the development of disease-specific atlases, e.g. for breast cancer, allows the identification of distinct disease subtypes. In this sense, the stratification of patients into different subgroups enables the application of personalized treatments [12].

Therefore, given the significant interest in the application of single-cell clustering for biomedical research, the field is experiencing rapid growth. Clustering methods applied to single-cell data are based on the foundation that cells belonging to a specific population exhibit a similar transcriptomic profile distinct from other cells within the biological sample. In recent times, different clustering algorithms have been developed to discern between distinct cell populations in scRNA-seq data. This is due to the fact that a comprehensive understanding of cell relationships provides invaluable insights into dynamic processes such as disease evolution and development [1]. Moreover, the discovered clusters form the basis for a majority of downstream analyses, highlighting the indispensability of a suitable clustering strategy. Considering the current research efforts in the field of scRNA-seq clustering analysis, our focus will be put on the use of multi-objective evolutionary algorithms [3,4].

A Multi-Objective Genetic Algorithm (MOGA) for cluster analysis of scRNA-seq data was proposed in [13]. The aim of this method is to optimize two objectives: intra-cluster distances and inter-cluster distances. The scRNA-seq datasets undergo dimensionality reduction to produce two-dimensional datasets which serve as input to the algorithm. The number of clusters remains constant throughout the optimization process, and this parameter is automatically set to the number of clusters detected in the data by the PhenoGraph method [14].

Alternatively, an Evolutionary Multi-objective Ensemble Pruning (EMEP) algorithm was presented in [15]. This method aims to optimize three objectives: the overall clustering deviation, within-cluster compactness, and the number of basic partition clusters. EMEP undertakes the clustering analysis as follows. First, unsupervised dimensionality reduction is applied to high-dimensional scRNA-seq data, projecting it into low-dimensional subspaces. Subsequently, three distinct clustering algorithms are applied to the reduced data, generating different clustering results and producing cluster ensembles.

Similarly to these two methods, the evolutionary algorithm proposed in this work also performs dimensionality reduction to decrease the dimensionality of scRNA-seq datasets. However, RVEA-CAST employs domain knowledge by using problem-aware mutation operators that dynamically change the number of clusters in the solution while improving each objective. This strategy allows us to efficiently optimize the objectives to obtain competitive clustering solutions without requiring the use of clustering ensembles.

3. Problem definition

The purpose of applying cluster analysis to single-cell transcriptomes is to discover different populations of cells, ensuring that cells belonging to the same population share a similar gene expression pattern, while having a distinct expression pattern from cells in other populations. This problem involves the processing of a dataset containing the expression levels of N cells across M genes. Therefore, the goal of the clustering task is to define K different cell clusters containing those input cells that show similar expression values for the considered genes.

In this work, we address the scRNA-seq clustering problem considering three objective functions. The first objective is to minimize the clustering deviation, which is determined by the total sum of distances between each cell and its corresponding cluster centroid. The clustering deviation is calculated as shown in Eq. (1):

$$\text{Clustering Deviation} = \sum_{k=1}^K \sum_{x_i \in \text{cluster}_k} d(x_i, c_k), \quad (1)$$

where K represents the number of clusters, c_k the centroid of the k th cluster (cluster_k), x_i the i th cell belonging to the cluster_k , and $d(x_i, c_k)$ the Euclidean distance between cell x_i and its belonging cluster centroid c_k .

The second objective is to optimize the compactness of the clustering, which measures how tightly packed the cells are within their respective clusters. Clustering compactness can be calculated as illustrated in Eq. (2). Therefore, this objective also has to be minimized:

$$\text{Clustering Compactness} = \frac{1}{N} \sum_{k=1}^K n_k \left(\frac{\sum_{x_i, x_j \in \text{cluster}_k} d(x_i, x_j)}{n_k(n_k - 1)/2} \right), \quad (2)$$

where N is the total number of cells in the dataset, K the number of clusters, n_k the number of cells in the cluster_k , and $d(x_i, x_j)$ the Euclidean distance between a pair of cells belonging to the same cluster.

The third objective to optimize is the Davies–Bouldin index (DBI), which is a clustering quality measure computed as the average similarity of each cluster with its most similar cluster. Such similarity is measured as the ratio of the intracluster distances to the intercluster distances [16]. The minimum score for this objective is zero, with lower scores indicating better clustering results. Therefore, it also needs to be minimized. Eq. (3) shows the calculation of DBI:

$$\text{DBI} = \frac{1}{K} \sum_{k=1}^K \max_{l \neq k} \left\{ \frac{S_k + S_l}{d(c_k, c_l)} \right\}, \text{ where } S_k = \frac{1}{n_k} \sum_{x_i \in \text{cluster}_k} d(x_i, c_k), \quad (3)$$

where S_k represents the average Euclidean distance of the cells in the k th cluster to the centroid c_k , n_k the number of cells in the cluster_k , K the number of clusters, and $d(c_k, c_l)$ the Euclidean distance between the cluster centroids c_k and c_l .

4. Proposed method

This section describes the approach proposed to conduct scRNA-seq clustering, detailing the algorithmic strategies and techniques defined in RVEA-CAST.

4.1. Data dimensionality reduction

The curse of dimensionality is inextricably linked to scRNA-seq data. This is due to the expression measurement of a vast number of transcripts to profile the gene expression patterns of the large number of cells within a biological sample. To address this challenge, dimensionality reduction techniques are essential to transform the original high-dimensional data into lower dimensional data. In this work, Principal Component Analysis (PCA) [17] is employed to reduce the large amount of gene expressions to 10 principal components per

cell, which collectively explain a large portion of the variability in the datasets. Subsequently, these 10 principal components undergo further reduction employing the Uniform Manifold Approximation and Projection (UMAP) technique [18] to obtain two-dimensional datasets (two coordinates per cell), which are then utilized in the subsequent cluster analysis.

4.2. Mutation operators

The mutation operators allow the generation of new candidate solutions, thus representing the main drivers of the evolutionary process in RVEA-CAST. Herein, a candidate solution is encoded as a vector $[c_1, c_2, \dots, c_K]$, where c_k denotes the centroid of the k th cell cluster, being each centroid represented by its two UMAP coordinates. To generate new candidate solutions according to the optimization goals of this problem, different mutation operators are defined, namely: (1) random mutation of one cluster centroid, (2) cluster split mutation for clustering deviation, (3) cluster split mutation for clustering compactness, and (4) cluster merging mutation for DBI. Notably, every devised operator acts on an input solution, performing random or problem-aware modifications to generate a mutated output solution. After that, the centroids are recalculated by averaging the coordinates of the cells belonging to each cluster.

The pseudocode in Algorithm 1 outlines the implementation of the random mutation. Initially, the algorithm randomly selects a centroid from the input solution (line 1 in Algorithm 1). Subsequently, the selected centroid is replaced by a new one, which is a randomly chosen cell from the data (lines 2–3 in Algorithm 1).

Algorithm 1 Pseudocode of the random mutation.

Input: *inSol*: input solution, *data*: reduced single-cell dataset.

Output: *outSol*: output solution.

- 1: *centroidToReplace* $\leftarrow$ *getRandomCentroid(inSol)*
 - 2: *newCentroid* $\leftarrow$ *selectRandomCell(data)*
 - 3: *outSol* $\leftarrow$ *replaceCentroid(inSol, centroidToReplace, newCentroid)*
 - 4: **return** *outSol*
-

The first of the problem-aware mutations is the split mutation aimed at improving the clustering deviation, illustrated in Algorithm 2. This mutation involves computing the clustering deviation of each individual cluster in the input solution (line 1 in Algorithm 2). Subsequently, the centroid of the cluster with maximum deviation is selected to be split (line 2 in Algorithm 2). To accomplish the split, the cell within that cluster that is farthest from the centroid is identified (line 3 in Algorithm 2). This selected cell becomes a new cluster centroid, leading to the generation of the output solution, where the cluster with maximum deviation is split (line 4 in Algorithm 2). A visual representation of the operator's mode of action is provided in Fig. 1.

Algorithm 2 Pseudocode of the split mutation for clustering deviation.

Input: *inSol*: input solution, *data*: reduced single-cell dataset.

Output: *outSol*: output solution.

- 1: *singleClustDev* $\leftarrow$ *computeSingleClustDev(inSol, data)*
 - 2: *centroidToSplit* $\leftarrow$ *getMaxClustDevCentroid(singleClustDev)*
 - 3: *farthestCell* $\leftarrow$ *getFarthestCell(data, centroidToSplit)*
 - 4: *outSol* $\leftarrow$ *addNewCentroid(inSol, farthestCell)*
 - 5: **return** *outSol*
-

The mutation aimed at improving the compactness of the clustering is also a split mutation, detailed in Algorithm 3. In this mutation operator, the clustering compactness is computed individually for each cluster in the input solution (line 1 in Algorithm 3). The centroid of the cluster with minimum compactness is then selected to be split (line 2 in Algorithm 3). Subsequently, the farthest pair of cells belonging to that cluster is identified to replace the old cluster centroid, completing

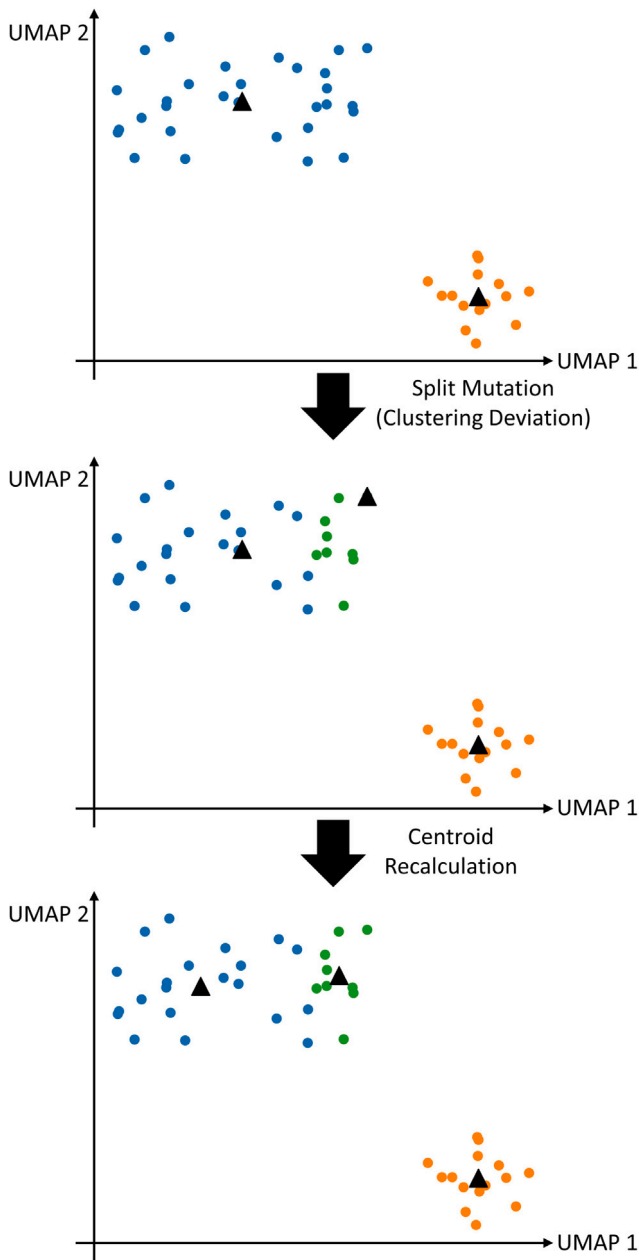


Fig. 1. Procedure of the split mutation aimed at improving the clustering deviation. The colors represent different clusters. The triangles indicate the positions of the centroids.

Algorithm 3 Pseudocode of the split mutation for clustering compactness.

Input: *inSol*: input solution, *data*: reduced single-cell dataset.

Output: *outSol*: output solution.

```

1: singleCompactness  $\leftarrow$  computeSingleCompactness(inSol, data)
2: centroidToSplit  $\leftarrow$  getMinCompactnessCentroid(singleCompactness)
3: farthestCellPair  $\leftarrow$  getFarthestCellPair(data, centroidToSplit)
4: outSol  $\leftarrow$  replaceCentroid(inSol, centroidToSplit, farthestCellPair)
5: return outSol

```

the splitting process (lines 3–4 in Algorithm 3). A visual representation of this operator's action is depicted in Fig. 2.

Lastly, the mutation aimed at improving the DBI is a merging mutation, which is detailed in Algorithm 4. This mutation identifies

the nearest centroid pair in the input solution (line 1 in Algorithm 4). The merged centroid is then calculated as the mean of each coordinate of the centroid pair (line 2 in Algorithm 4). Finally, the pair of nearest centroids is replaced by the merged centroid, completing the merging process and generating the output solution (line 3 in Algorithm 4). The operation performed by this mutation operator is represented in Fig. 3.

Algorithm 4 Pseudocode of the merging mutation for DBI.

Input: *inSol*: input solution, *data*: reduced single-cell dataset.

Output: *outSol*: output solution.

```

1: nearestCentPair  $\leftarrow$  getNearestCentroids(inSol, data)
2: mergedCentroid  $\leftarrow$  getMergedCentroid(nearestCentPair)
3: outSol  $\leftarrow$  replaceCentroid(inSol, nearestCentPair, mergedCentroid)
4: return outSol

```

It can be observed that our algorithm dynamically modifies the number K of clusters in a clustering solution through problem-aware mutation operators. To ensure controlled variability, the scope of the number of clusters is limited to a range of values $minK$ – $maxK$ based on the estimation of the number of clusters in the input dataset using the Bayesian Information Criterion (BIC). BIC is a statistical measure used for model selection, providing a quantitative score where lower values indicate a better model fit [19].

The random mutation operator and the problem-aware mutation operators are employed in the mutation procedure of the proposed approach, whose implementation is detailed in Algorithm 5. This procedure is applied as follows.

Algorithm 5 Pseudocode of the mutation procedure.

Input: *inSol*: input solution, *data*: reduced single-cell dataset, K : number of clusters in the input solution, $minK$: minimum number of clusters, $maxK$: maximum number of clusters.

Output: *outSol*: output solution.

```

1: randomNum  $\leftarrow$  rand()
2: if  $K == minK$  then
3:   switch (randomNum%3)
4:     case 0: outSol  $\leftarrow$  randomMutation(inSol, data)
5:     case 1: outSol  $\leftarrow$  splitClustDevMutation(inSol, data)
6:     case 2: outSol  $\leftarrow$  splitClustCompactnessMutation(inSol, data)
7:   end switch
8: else if  $K == maxK$  then
9:   switch (randomNum%2)
10:    case 0: outSol  $\leftarrow$  randomMutation(inSol, data)
11:    case 1: outSol  $\leftarrow$  mergingMutation(inSol, data)
12:   end switch
13: else
14:   switch (randomNum%4)
15:     case 0: outSol  $\leftarrow$  randomMutation(inSol, data)
16:     case 1: outSol  $\leftarrow$  splitClustDevMutation(inSol, data)
17:     case 2: outSol  $\leftarrow$  splitClustCompactnessMutation(inSol, data)
18:     case 3: outSol  $\leftarrow$  mergingMutation(inSol, data)
19:   end switch
20: end if
21: outSol  $\leftarrow$  centroidRecalculation(outSol)
22: return outSol

```

If the input solution has the minimum number of clusters allowed ($minK$), the merging mutation cannot be applied. In this scenario, only the random and split mutations are available, and one of them is randomly chosen (lines 2–7 in Algorithm 5). Conversely, if the input solution has the maximum number of clusters allowed ($maxK$), the split mutation cannot be applied. Thus, only the random and merging mutations are available, and one of them is randomly selected (lines 8–12 in Algorithm 5). In all other cases, where the input solution is neither at the minimum nor maximum number of clusters, all four mutations are available, and one of them is randomly chosen (lines

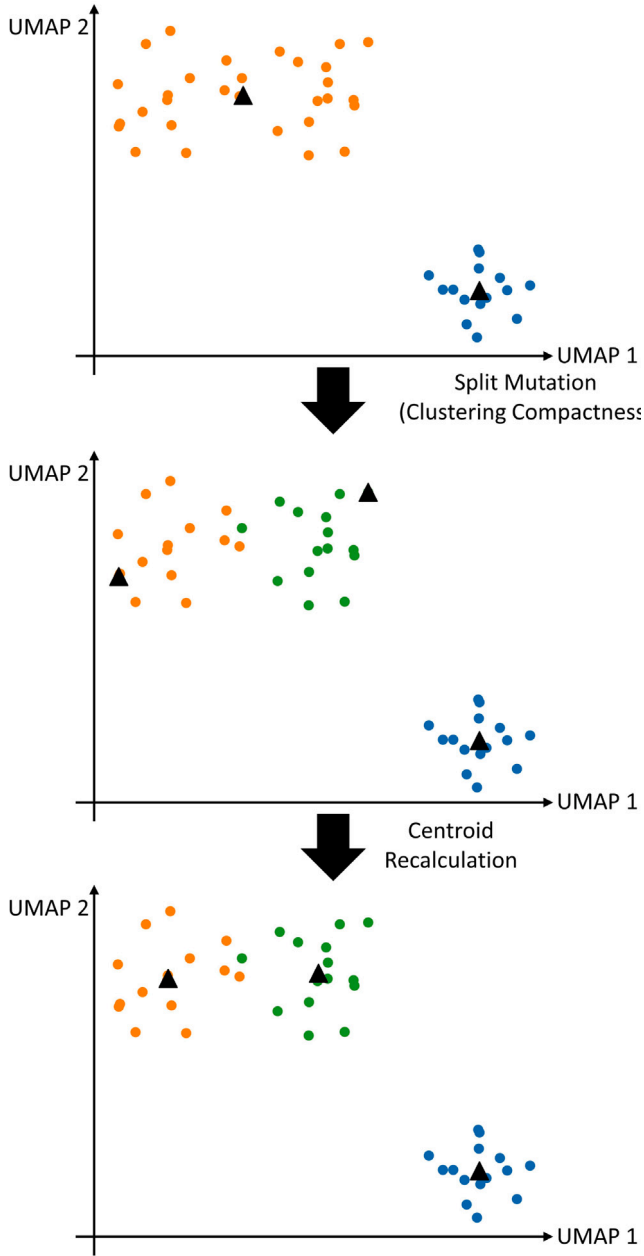


Fig. 2. Procedure of the split mutation aimed at improving the clustering compactness. The colors represent different clusters. The triangles indicate the positions of the centroids.

13–19 in Algorithm 5). As a final step, the centroids of the mutated solution are updated by averaging the coordinates of the cells from each cluster (line 21 in Algorithm 5).

A flowchart illustrating the mutation procedure in RVEA-CAST is depicted in Fig. 4.

4.3. Proposed algorithm: RVEA-CAST

In order to optimize clustering solutions, we propose a Reference Vector-guided Evolutionary Algorithm for Cluster Analysis of Single-cell Transcriptomes (RVEA-CAST). This algorithm builds upon the Reference Vector-guided Evolutionary Algorithm (RVEA) introduced in [20]. This evolutionary approach adopts the idea of using reference vectors to split the objective space of a multi-objective optimization

problem into different partitions or subspaces. In this way, the processing of each particular subspace can be effectively handled to boost the overall optimization process. In addition, RVEA employs a scalarization approach, designated as angle-penalized distance (APD), to pursue a balance between convergence and diversity in the defined subspaces for solution selection purposes. Furthermore, the distribution of the reference vectors is dynamically adjusted based on the scales of the objective functions using an adaptation strategy.

The main structure of the proposed RVEA-CAST is shown in Algorithm 6. The algorithm takes as input the population size ($popSize$), the maximum number of generations considered as stop criterion ($maxGen$), the initial reference vectors ($refV_0$), the minimum and maximum number of clusters ($minK$ and $maxK$), and the reduced single-cell dataset ($data$). The output is the final population of solutions found (P_{maxGen}).

Algorithm 6 Pseudocode of RVEA-CAST.

Input: $popSize$: population size, $maxGen$: maximum number of generations, $refV_0$: initial reference vectors, $minK$: minimum number of clusters, $maxK$: maximum number of clusters, $data$: reduced single-cell dataset.

Output: P_{maxGen} : final population.

```

1:  $g \leftarrow 0$ 
2:  $P_g \leftarrow \text{initRandomPop}(popSize, data, minK, maxK)$ 
3: while  $g < maxGen$  do
4:    $Q_g \leftarrow \emptyset$ 
5:   for  $i \leftarrow 1$  to  $popSize$  do
6:      $mutSol \leftarrow \text{mutationProcedure}(P_{g,i}, data, P_{g,i}.K, minK, maxK)$ 
7:      $Q_g \leftarrow Q_g \cup mutSol$ 
8:   end for
9:    $P_g \leftarrow P_g \cup Q_g$ 
10:   $P'_g \leftarrow \text{objectiveValueTranslation}(P_g)$ 
11:   $\bar{P}_g \leftarrow \text{populationPartitioning}(P'_g, refV_g)$ 
12:   $d_g \leftarrow \text{APDCalculation}(\bar{P}_g, refV_g)$ 
13:   $P_{g+1} \leftarrow \text{elitismSelection}(\bar{P}_g, d_g)$ 
14:   $refV_{g+1} \leftarrow \text{referenceVectorAdaption}(g, P_{g+1}, refV_g, refV_0)$ 
15:   $g \leftarrow g + 1$ 
16: end while
17: return  $P_{maxGen}$ 

```

Initially, the population of $popSize$ individuals is initialized by randomly selecting K cells from the $data$ as cluster centroids, where K is a value between $minK$ and $maxK$, both included (line 2 in Algorithm 6). Then, the iterative process of the algorithm begins (line 3 in Algorithm 6), covering each generation g until the stopping criterion ($maxGen$) is reached. In each of the generations constituting this iterative process, the mutation procedure is firstly applied to the individuals from the population P_g , generating the offspring population of mutated solutions Q_g (lines 4–8 in Algorithm 6). The offspring population is then combined with the parent population before undergoing the selection of the most promising individuals for the next generation (line 9 in Algorithm 6). This selection is guided by the reference vectors and follows 4 key steps:

1. **Objective value translation:** in a first step, the objective values of the solutions in P_g are transformed into P'_g using the minimum objective values from P_g (line 10 in Algorithm 6).
2. **Population partitioning:** the objective-translated population P'_g is partitioned into $|refV|$ subpopulations, associating each individual with its closest reference vector based on the acute angle between them (line 11 in Algorithm 6).
3. **APD calculation:** in this step, the APD of each solution in the subpopulations is calculated. The use of APD as a selection criterion allows balancing convergence and diversity properties within the subpopulation space (line 12 in Algorithm 6).

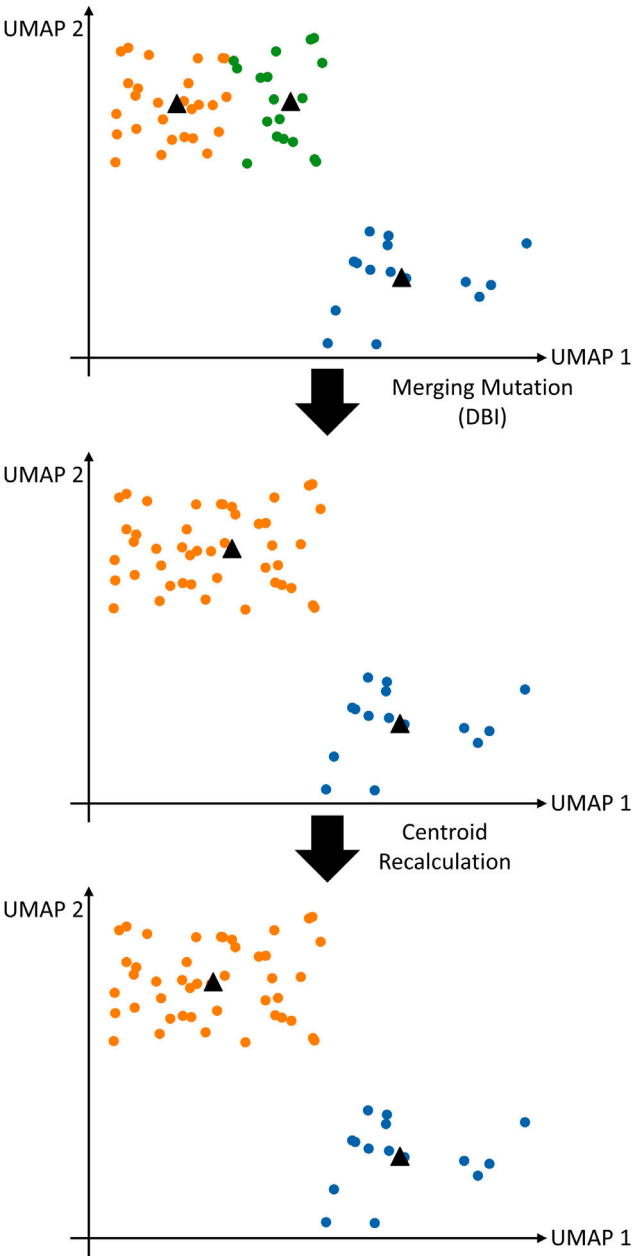


Fig. 3. Procedure of the merging mutation aimed at improving the DBI of the clustering. The colors represent different clusters. The triangles indicate the positions of the centroids.

4. *Elitism selection*: after computing the APD values, elitism-based selection is performed by choosing the individuals with the minimum APD in each subpopulation. The selected individuals are employed to define the population of the next generation P_{g+1} (line 13 in Algorithm 6).

Following the elitism selection, the reference vectors are updated based on the ranges of the objective values of the P_{g+1} population to boost the search for uniformly distributed solutions, in accordance with the status of the optimization process (line 14 in Algorithm 6). The algorithm performs its main loop until $maxGen$ iterations are reached, at which point the final population of solutions is returned as output (line 17 in Algorithm 6).

A flowchart depicting the steps of the RVEA-CAST approach is provided in Fig. 5.

Table 1
Characteristics of the single-cell RNA-seq datasets employed in this work. The number of cells, genes, and cell types are shown for each dataset. Datasets are sorted by number of cell types in ascending order.

Dataset	Cells	Genes	Cell types
Ginhoux [22]	251	11 834	3
Buettner [23]	182	8989	3
Ting [24]	114	14 405	5
Treutlein [25]	80	959	5
Deng [26]	135	12 548	7
Pollen [27]	249	14 805	11
Macosko [28]	6418	12 822	39
Zeisel [29]	3005	4412	48
Tasic [30]	1727	5832	48
Manno [31]	4029	20 058	56

5. Results

This section presents the results obtained by the proposed RVEA-CAST and compares them with those from other methods of the field. For this purpose, we first introduce the datasets, experimental settings, and evaluation metrics employed in this study. Afterwards, the comparative evaluation of RVEA-CAST with other methods is provided. Finally, a biological analysis of the obtained results is presented.

5.1. Datasets

Ten real scRNA-seq datasets were herein employed to assess the performance of RVEA-CAST against other methods proposed in the scientific literature. These datasets, listed in Table 1, were obtained from [21] and preprocessed as detailed in its supplementary material.

A key factor in dataset selection was the availability of validated cell identities, which are essential for calculating clustering evaluation metrics. Additionally, the datasets cover a wide range of problem sizes and biological scenarios, ensuring a robust evaluation of RVEA-CAST. Specifically, they vary in the number of cells (from 80 to 6418), genes (from 959 to 20058), and cell types (from 3 to 56), as detailed in Table 1.

A brief description of each dataset is provided as follows:

- The Ginhoux dataset profiles dentritic cell progenitors at distinct differentiation stages [22].
- The Buettner dataset is composed of mouse embrionic stem cells on different cell cycle phases [23].
- The Ting dataset contains different subpopulations of pancreatic circulating tumor cells [24].
- The Treutlein dataset measures the transcriptomic profiles of cells at different stages of alveolar differentiation [25].
- The Deng dataset is composed of cells isolated from mouse preimplantation embryos at distinct differentiation stages [26].
- The Pollen dataset contains different subpopulations of cells, including neural cells, blood cells, and dermal cells, among others [27].
- The Macosko dataset is composed of distinct populations of mouse retinal cells [28].
- The Zeisel dataset measures the transcriptomic profiles of different cell subpopulations from the mouse somatosensory cortex and hippocampal CA1 region [29].
- The Tasic dataset is composed of different transcriptomic cell types from the primary visual cortex [30].
- The Manno dataset contains different cell types in human ventral midbrain development, including dopaminergic neurons, which are relevant for Parkinson's disease [31].

From a biological perspective, the selected datasets cover a broad spectrum of biological contexts. Specifically, the Ginhoux, Deng, Treutlein, and Manno datasets capture distinct developmental and differentiation stages, evaluating the algorithm's ability to distinguish transitions

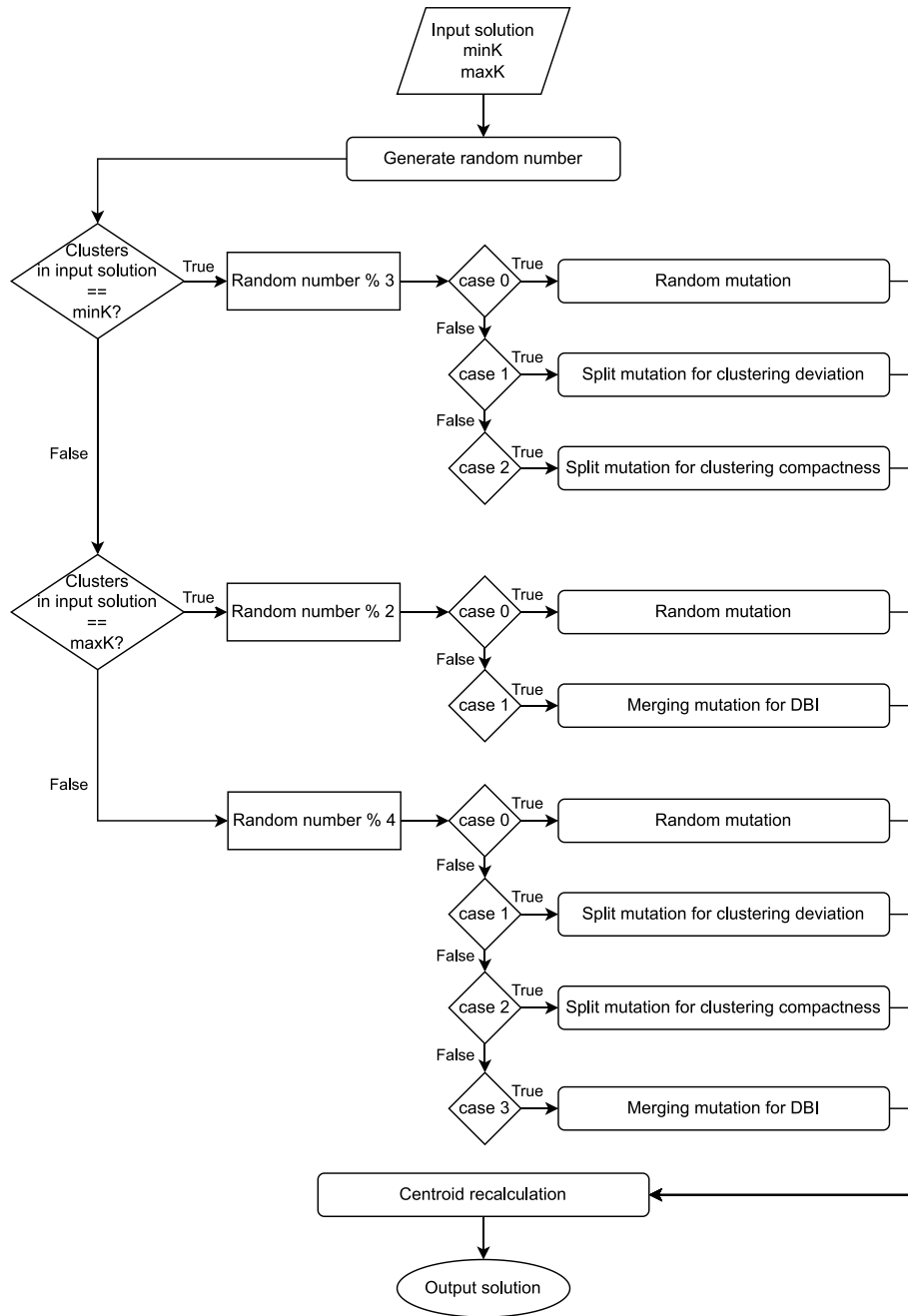


Fig. 4. Flowchart of the mutation procedure of RVEA-CAST.

between cell states. The Buettner dataset assesses the algorithm's performance in clustering cells at different cell cycle phases. The Ting dataset evaluates the algorithm's ability to distinguish heterogeneous tumor cell populations. The Pollen dataset tests the algorithm's capacity to differentiate tissue-specific cell populations. Finally, the Macosko, Zeisel, and Tasic datasets assess the algorithm's effectiveness in identifying cell types within neural tissues, where transcriptional differences between cell types are often subtle.

These datasets collectively provide the means to conduct a comprehensive and robust evaluation, ensuring that clustering algorithms are tested across a diverse range of biological conditions.

5.2. Experimental settings

In this experimental evaluation, RVEA-CAST is compared with other multi-objective methods previously proposed for scRNA-seq clustering:

MOGA [13] and EMEP [15]. Table 2 presents the experimental settings of RVEA-CAST and the competing methods. The number of evaluations for our method and MOGA was set to 1000, matching the number of evaluations in EMEP to ensure a fair comparison. To determine the optimal values for the input parameters of our method, a parametric study was conducted, leading to the selection of a population size of 50 individuals and a number of generations of 20.

In the clustering field, one of the main challenges to tackle lies in selecting the optimal number of clusters. In fact, some methods employ the real number of clusters from the labels, which is unavailable in most cases. To address this issue, our method estimates the number of clusters using BIC over Gaussian Mixture models in a range from 2 to 50 clusters. To ensure a fair comparison, those methods that previously employed the real number of clusters from the labels have been updated to adopt this strategy.

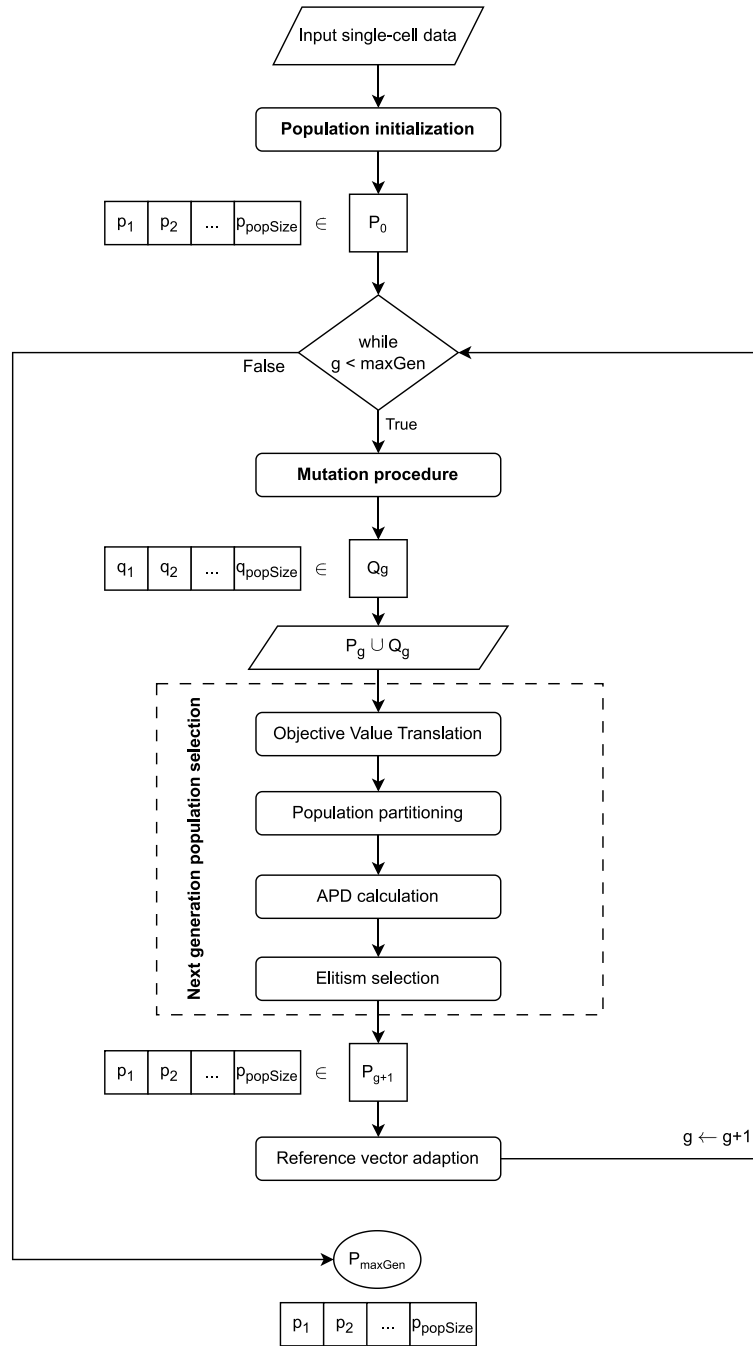


Fig. 5. Flowchart of the RVEA-CAST algorithm.

Table 2

Information and experimental settings of the different methods. Population size (P), number of generations (G), and number of evaluations (E) are shown.

Method	Clustering type	Evolutionary algorithm	P	G	E	Independent runs
RVEA-CAST	Centroid-based	RVEA	50	20	1000	31
MOGA [13]	Centroid-based	MOGA	50	20	1000	31
EMEP [15]	Ensemble clustering	MOEA/D	–	–	1000	31

To promote the attainment of well-distributed sets of solutions, we employed the Das–Dennis method [32] to generate a set of reference points. These reference points are used to define the reference vectors required by RVEA-CAST. Eq. (4) was applied to compute the number H of reference points needed for this purpose:

$$H = \binom{O+p-1}{p}, \quad (4)$$

where O denotes the number of objectives to optimize and p the number of partitions. The number of partitions p must be greater than the number of objectives which is $O = 3$. Furthermore, H should be close

but not greater than the population size. Therefore, $p = 8$ partitions were selected to set the number of reference points to 45.

To obtain statistically robust results due to the stochastic behavior of evolutionary algorithms, 31 independent executions for each experiment were run. To ensure the collection of solutions within a practical time period, the computing time per execution was limited to six hours.

For the statistical analysis of the results, the Kolmogorov–Smirnov test was used to assess normality, and the Levene test was used to evaluate the homogeneity of variances. If both assumptions were met, the ANOVA parametric test was employed to determine whether there were statistically significant differences between the algorithms. If any of the assumptions was violated, the non-parametric Mann–Whitney U test was employed instead. The significance level for all tests was set to 0.05.

All the experiments were conducted on a computer with an Intel Core i7-8700 CPU at 3.20 GHz and a RAM of 32 GB, whose operating system is Windows 10 Pro. All the software was developed using Python 3.10.6, the IDE Visual Studio Code 1.80, and the packages pymoo (v0.6.0) [33] and Scikit-learn (v1.2.0) [34].

5.3. Evaluation metrics

Two evaluation metrics are adopted in this comparative study to verify the relevance of the proposed RVEA-CAST. In particular, we employ the Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) due to their widespread presence in the cluster analysis of single-cell transcriptomes.

NMI quantifies the similarity between two clustering partitions by assessing the mutual information they share while accounting for entropy of the individual clusters [35]. It provides a normalized score between 0 and 1, where higher values indicate a stronger concordance between both partitions. NMI can be calculated as shown in Eq. (5):

$$NMI = \frac{2 \times I(y, y')}{H(y) + H(y')}, \quad (5)$$

where y and y' represent the real and the predicted identities, respectively. $I(y, y')$ is the mutual information between y and y' , and H the entropy.

ARI is a measure of the similarity between two clustering solutions which relies on a pair-wise comparison between both solutions while accounting for chance agreement [36]. The overlap between these two solutions can be presented in a contingency table. In this table, each entry indicates the number of cells in agreement with both partitions. ARI provides a normalized score between -1 and 1 , where higher values indicate stronger agreement between the clustering solutions. Eq. (6) shows the calculation of ARI:

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - \sum_i \binom{a_i}{2} \cdot \sum_j \binom{b_j}{2} / \binom{N}{2}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \sum_i \binom{a_i}{2} \cdot \sum_j \binom{b_j}{2} / \binom{N}{2}}, \quad (6)$$

where N is the total number of cells, n_{ij} the number of cells in common between the i th cluster of labels and the j th cluster of predictions, a_i the number of cells in common between the i th cluster of labels and the predictions, and b_j the number of cells in common between the j th cluster of predictions and the labels.

5.4. Comparative evaluation

The results of RVEA-CAST and the state-of-the-art methods MOGA [13] and EMEP [15], and the results of the statistical tests comparing RVEA-CAST to these methods are presented in Tables 3 and 4 for NMI, and Tables 5 and 6 for ARI.

The absence of results for EMEP on the datasets Macosko, Zeisel, Tasic, and Manno is due to the algorithm's extensive computing time as a result of the use of cluster ensembles, exceeding the limit of six hours per execution. In contrast, both RVEA-CAST and MOGA were able

Table 3

NMI values of the different methods after 31 independent runs. Values are shown in format *median*_{±quartile deviation}. The best NMI values are highlighted in bold. The dashes indicate that the results are not available.

Dataset	RVEA-CAST	MOGA [13]	EMEP [15]
Ginhoux	0.51 _{±0.01}	0.42 _{±0.04}	0.46 _{±0.03}
Buettner	0.76 _{±0.00}	0.73 _{±0.02}	0.63 _{±0.10}
Ting	0.91 _{±0.00}	0.73 _{±0.04}	0.93 _{±0.13}
Treutlein	0.81 _{±0.00}	0.71 _{±0.01}	0.75 _{±0.06}
Deng	0.79 _{±0.02}	0.57 _{±0.00}	0.74 _{±0.06}
Pollen	0.95 _{±0.01}	0.57 _{±0.02}	0.93 _{±0.13}
Macosko	0.49 _{±0.00}	0.52 _{±0.02}	–
Zeisel	0.54 _{±0.00}	0.50 _{±0.00}	–
Tasic	0.44 _{±0.00}	0.30 _{±0.02}	–
Manno	0.53 _{±0.00}	0.47 _{±0.03}	–

Table 4

Results of statistical tests comparing RVEA-CAST with the state-of-the-art methods in terms of NMI. Statistically significant differences are highlighted in bold. The dashes indicate that the results are not available.

Dataset	RVEA-CAST vs MOGA [13]	RVEA-CAST vs EMEP [15]
Ginhoux	<0.001	0.049
Buettner	<0.001	0.003
Ting	<0.001	0.488
Treutlein	<0.001	<0.001
Deng	<0.001	<0.001
Pollen	<0.001	0.019
Macosko	<0.001	–
Zeisel	<0.001	–
Tasic	<0.001	–
Manno	<0.001	–

Table 5

ARI values of the different methods after 31 independent runs. Values are shown in format *median*_{±quartile deviation}. The best ARI values are highlighted in bold. The dashes indicate that the results are not available.

Dataset	RVEA-CAST	MOGA [13]	EMEP [15]
Ginhoux	0.54 _{±0.01}	0.33 _{±0.08}	0.48 _{±0.03}
Buettner	0.78 _{±0.01}	0.77 _{±0.04}	0.46 _{±0.18}
Ting	0.89 _{±0.00}	0.61 _{±0.12}	0.88 _{±0.21}
Treutlein	0.86 _{±0.00}	0.73 _{±0.01}	0.73 _{±0.09}
Deng	0.58 _{±0.03}	0.36 _{±0.00}	0.51 _{±0.09}
Pollen	0.94 _{±0.01}	0.26 _{±0.03}	0.85 _{±0.29}
Macosko	0.23 _{±0.00}	0.30 _{±0.06}	–
Zeisel	0.25 _{±0.00}	0.25 _{±0.03}	–
Tasic	0.14 _{±0.00}	0.07 _{±0.01}	–
Manno	0.23 _{±0.00}	0.18 _{±0.02}	–

to efficiently handle these complex datasets and produce meaningful results in reasonable time.

In terms of NMI, RVEA-CAST outperforms state-of-the-art methods in eight out of ten datasets (Table 3). In these datasets, the NMI values attained by RVEA-CAST present statistically significant differences over all competing methods, showcasing a remarkable enhancement in clustering performance (Table 4). This improvement is prominent in the Buettner dataset, where RVEA-CAST achieves an NMI of 0.76, representing a 20.6% improvement over EMEP. Similarly, in the Pollen dataset, RVEA-CAST attains an NMI of 0.95, showing a 66.7% improvement over MOGA. In the complex Tasic dataset, where only MOGA could produce results, RVEA-CAST achieves an NMI improvement of 46.7%.

Regarding the comparison in terms of ARI, RVEA-CAST attains the highest ARI values in nine out of ten datasets (Table 5). Among the six datasets where results from both state-of-the-art methods are available, our approach achieves statistically significant improvements in 11 out of 12 comparisons. Furthermore, RVEA-CAST significantly outperforms MOGA in two of the most complex datasets: Tasic and Manno (Table 6). The Pollen dataset also highlights the relevance of our method,

Table 6

Results of statistical tests comparing RVEA-CAST with the state-of-the-art methods in terms of ARI. Statistically significant differences are highlighted in bold. The dashes indicate that the results are not available.

Dataset	RVEA-CAST vs MOGA [13]	RVEA-CAST vs EMEP [15]
Ginhoux	<0.001	<0.001
Buettner	<0.001	0.033
Ting	<0.001	0.473
Treutlein	<0.001	<0.001
Deng	<0.001	<0.001
Pollen	<0.001	0.001
Macosko	<0.001	–
Zeisel	0.937	–
Tasic	<0.001	–
Manno	<0.001	–

where RVEA-CAST obtains an ARI of 0.94, representing a remarkable improvement of 261.5% over MOGA. In comparison with EMEP, the Buettner dataset reveals that our approach accomplishes a noteworthy improvement of 69.6% over EMEP.

Overall, these results indicate that RVEA-CAST consistently outperforms state-of-the-art methods in both NMI and ARI metrics. Specifically, if NMI and ARI values are analyzed together, RVEA-CAST achieves non-dominated solutions in most datasets (nine out of ten). In these datasets where non-dominated solutions are obtained, RVEA-CAST reports the best solutions in eight out of nine datasets. These results indicate that the clusters produced by RVEA-CAST exhibit a higher degree of similarity to the ground-truth clusters, leading to more accurate clustering solutions.

Furthermore, RVEA-CAST showcases robustness by consistently obtaining high-quality results across multiple runs, as it can be noticed from the low quartile deviation of the results. The capabilities of our method to generate clusters that align closely with the true cluster labels denote its effectiveness in capturing the underlying structure of scRNA-seq data. Moreover, RVEA-CAST shows a stable behavior on different optimization scenarios by dynamically modifying the number of clusters in the solutions, adapting the search according to the characteristics of the problem instance under study.

5.5. Importance of the intelligent mutation operators

To evaluate the impact of the devised intelligent mutation operators on RVEA-CAST's performance, we conducted ablation experiments on the datasets under study. Specifically, we compared RVEA-CAST with a variant that excludes these intelligent mutation operators (RVEA-CAST-w/o-IMO).

To this end, we employed the Inverted Generational Distance (IGD), a well-known multi-objective optimization performance metric, computed as detailed in Eq. (7). This metric assesses the quality of a Pareto approximation front with respect to the Pareto optimal front. Specifically, it measures the average Euclidean distance from every solution in the Pareto optimal front to its closest solution in the Pareto approximation front. Therefore, lower IGD values indicate a better Pareto approximation front.

$$IGD(A) = \frac{1}{|O|} \left(\sum_{i=1}^{|O|} \tilde{d}_i^2 \right)^{1/2}, \quad (7)$$

where A represents a Pareto approximation front, O is the Pareto optimal front, $|O|$ refers to the number of solutions in O , and $\tilde{d}_i$ denotes the Euclidean distance from the i th solution $o_i \in O$ to its closest solution in A .

Table 7 shows that RVEA-CAST achieves the lowest IGD values in nine out of ten cases. In eight of these cases, RVEA-CAST outperforms the version without the intelligent mutation operators. Notably,

Table 7

IGD values of RVEA-CAST and RVEA-CAST-w/o-IMO after 31 independent runs. Values are shown in format *median*_{±quartile deviation}. The best IGD values are highlighted in bold.

Dataset	RVEA-CAST	RVEA-CAST-w/o-IMO
Ginhoux	0.059 _{±0.007}	0.061 _{±0.005}
Buettner	0.072 _{±0.008}	0.072 _{±0.018}
Ting	0.024 _{±0.003}	0.026 _{±0.006}
Treutlein	0.073 _{±0.013}	0.027 _{±0.026}
Deng	0.099 _{±0.026}	0.258 _{±0.048}
Pollen	0.132 _{±0.023}	0.161 _{±0.027}
Macosko	0.130 _{±0.021}	0.202 _{±0.032}
Zeisel	0.179 _{±0.034}	0.199 _{±0.027}
Tasic	0.120 _{±0.017}	0.130 _{±0.037}
Manno	0.187 _{±0.047}	0.270 _{±0.075}

Table 8

Results of statistical tests comparing RVEA-CAST with RVEA-CAST-w/o-IMO in terms of IGD. Statistically significant differences are highlighted in bold.

Dataset	RVEA-CAST vs RVEA-CAST-w/o-IMO
Ginhoux	0.959
Buettner	0.704
Ting	0.215
Treutlein	<0.001
Deng	<0.001
Pollen	<0.001
Macosko	<0.001
Zeisel	0.038
Tasic	0.159
Manno	0.003

RVEA-CAST exhibits statistically significant improvements in half of the evaluated datasets (Table 8), with IGD improvements of up to 61.6% compared to RVEA-CAST-w/o-IMO. This result showcases the contribution of the intelligent mutation operators towards the attainment of better solutions in these scenarios.

The Distance to the Ideal Point (DIP) metric was also employed to measure quality based on a single solution from the entire Pareto front, as defined in Eq. (8). Specifically, this metric quantifies the minimum distance between solutions in the Pareto approximation front and the ideal reference point. Thus, lower DIP values indicate better quality.

$$DIP(A) = \min_{a_i \in A} \left(d^2(a_i, r) \right), \quad (8)$$

where A represents a Pareto approximation front, and $d^2(a_i, r)$ denotes the Euclidean distance from the i th solution $a_i \in A$ to the ideal reference point r , which is a hypothetical point resulting from the best observed values for each objective.

Table 9 indicates that RVEA-CAST attains the lowest DIP values in all experimental scenarios. Specifically, RVEA-CAST outperforms RVEA-CAST-w/o-IMO in 50% of the comparisons, all of which present statistically significant differences (Table 10). In particular, RVEA-CAST achieves performance improvements of up to 24.1% over the variant lacking the intelligent mutation operators.

Overall, the results of these ablation experiments highlight the usefulness of the intelligent mutation operators, as their inclusion leads to the attainment of the best results in practically all cases (19 out of 20 comparisons). Furthermore, statistically significant improvements over the version without intelligent mutation operators are observed in half of the datasets under study, particularly in the most challenging datasets, emphasizing the importance of these operators in complex clustering scenarios.

5.6. Scalability study

To assess the scalability of our algorithm, we analyze execution times across all datasets considered. Fig. 6 presents a scatter plot of

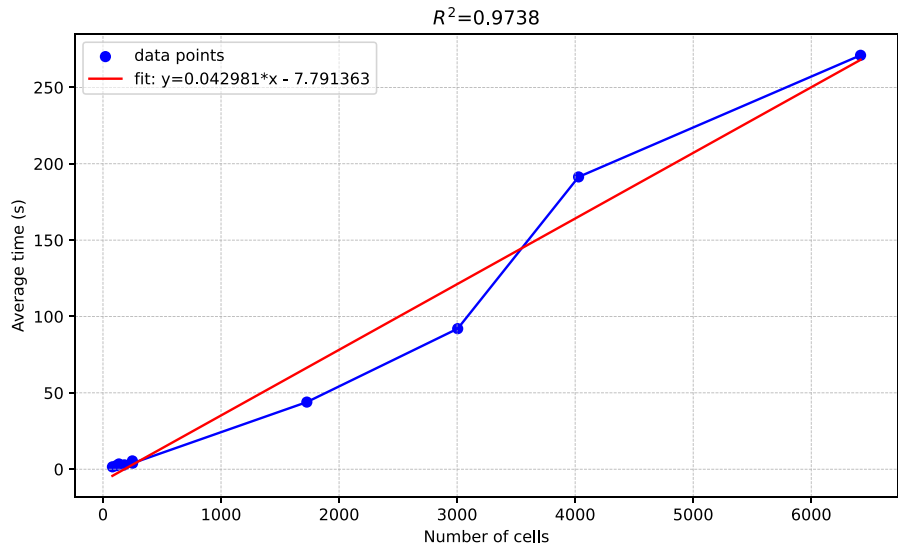


Fig. 6. Scatter plot of average execution times (in seconds) for the datasets under study. The fitted line (red) models the relationship between dataset size and computational time.

Table 9
DIP values of RVEA-CAST and RVEA-CAST-w/o-IMO after 31 independent runs. Values are shown in format *median*_{±quartile deviation}. The best DIP values are highlighted in bold.

Dataset	RVEA-CAST	RVEA-CAST-w/o-IMO
Ginhoux	0.150 _{±0.003}	0.150 _{±0.003}
Buettner	0.052 _{±0.001}	0.052 _{±0.001}
Ting	0.120 _{±0.000}	0.120 _{±0.000}
Treutlein	0.066 _{±0.004}	0.066 _{±0.002}
Deng	0.243 _{±0.011}	0.320 _{±0.049}
Pollen	0.130 _{±0.012}	0.169 _{±0.050}
Macosko	0.172 _{±0.018}	0.213 _{±0.036}
Zeisel	0.167 _{±0.011}	0.209 _{±0.025}
Tasic	0.209 _{±0.016}	0.209 _{±0.016}
Manno	0.215 _{±0.027}	0.243 _{±0.054}

Table 10
Results of statistical tests comparing RVEA-CAST with RVEA-CAST-w/o-IMO in terms of DIP. Statistically significant differences are highlighted in bold.

Dataset	RVEA-CAST vs RVEA-CAST-w/o-IMO
Ginhoux	0.687
Buettner	0.372
Ting	0.604
Treutlein	0.388
Deng	<0.001
Pollen	<0.001
Macosko	<0.001
Zeisel	<0.001
Tasic	0.302
Manno	0.023

the average execution times for these datasets, along with a fitted line modeling the relationship between dataset size and computational time.

Notably, execution times remain below one minute for datasets with around 2000 cells. Even for the largest dataset (6418 cells), execution time remains reasonably low at approximately 270 s.

Furthermore, the fitted line enables estimation of the algorithm’s performance on large-scale datasets. With an R^2 value of 97.38%, the model exhibits a strong correlation, providing high-confidence estimations of execution time for larger datasets.

Overall, while execution time increases with the number of cells, the linear trend indicates that the RVEA-CAST algorithm scales efficiently.

5.7. Biological analysis

To provide a comprehensive understanding of the biological relevance of our method, we conducted a biological analysis of the clustering results using four datasets, each representing distinct scenarios in terms of the number of cell types. Specifically, we selected the Ting dataset to represent a case with few of them (5 cell types), the Pollen dataset as an intermediate dataset (11 cell types), and the Zeisel and Manno datasets to depict scenarios with many of them (48 and 56 cell types, respectively). The cluster assignments of the solutions previously presented in Tables 3 and 5 for the Ting, Pollen, Zeisel, and Manno datasets in each method were compared with the real cluster identities. To achieve this, differential gene expression analysis was performed to detect the differentially expressed genes (DEGs) between clusters for each clustering solution using the FindAllMarkers function from the Seurat package (v4.3.0) [37]. For this purpose, the Wilcoxon Rank Sum test was employed, and the log fold-change was set to 0.25, following the default configuration. For the enrichment analysis of DEGs, clusterProfiler was used [38].

For the detection of DEGs, the significance level of the p-values adjusted for multiple testing was set to 0.01. To compare the DEGs from the real clusters with those from the clusters predicted by the different methods, Venn diagrams were used, as shown in Figs. 7, 8, 9, and 10, illustrating the overlap between the DEGs of each method and those from the real clusters. Accuracy and F1-score were computed for each comparison to provide a deeper insight [39]. To achieve this, the number of genes detected in both the real and predicted clusters were considered true positive cases, the number of genes detected in the predicted but not in the real clusters were considered false positive cases, and the number of genes detected in the real but not in the predicted clusters were considered false negative cases.

The Venn diagrams in Fig. 7 illustrate the overlap between the DEGs identified in the real clusters and those predicted by each method for the Ting dataset. It can be observed that the DEGs detected in the RVEA-CAST solution are highly similar to those detected from the real clusters (2688 matches out of 3009). It is noticeable that the two circles are concentric and have the same radius, indicating a high concordance (as shown in Fig. 7(a)). This idea is supported by the results obtained by our method in terms of accuracy and F1-score (80.55% and 89.23%, respectively), outperforming the results obtained by MOGA and EMEP (Figs. 7(b) and 7(c)).

The results of the biological analysis in the Pollen dataset are shown in Fig. 8. This dataset presents more cell types than the previous one. In

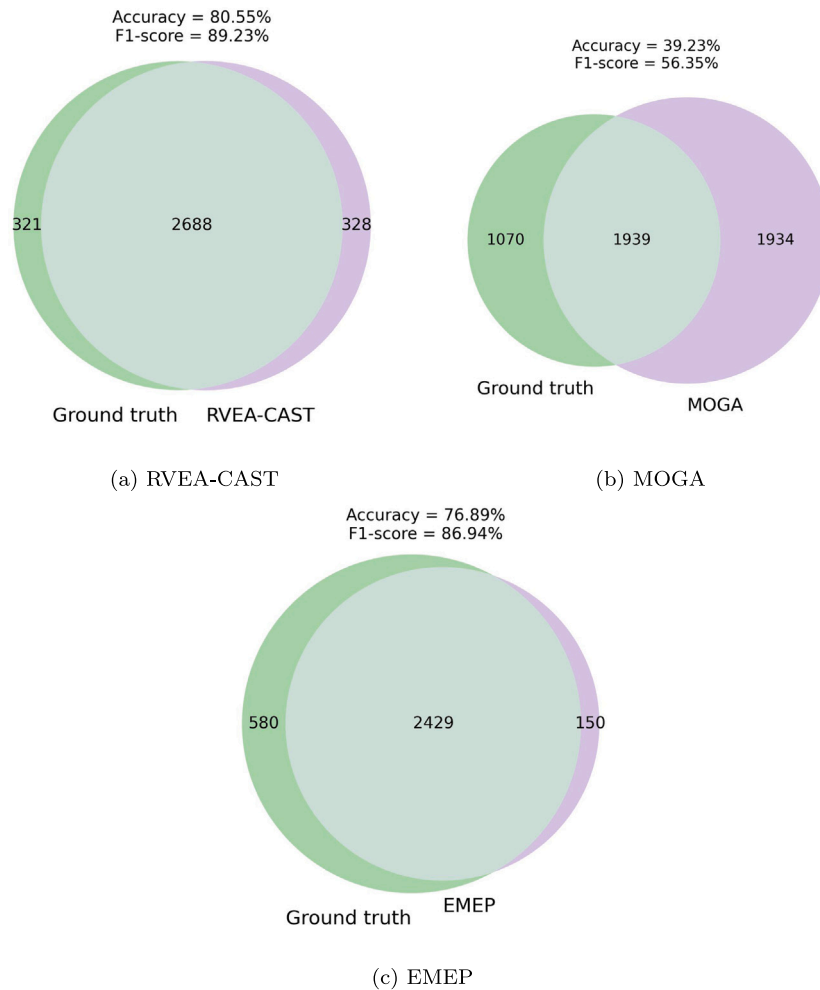


Fig. 7. Venn diagrams of the DEGs detected in the real clusters with those detected by RVEA-CAST (a), MOGA (b), and EMEP (c) in the Ting dataset. Accuracy and F1-score of each method are also shown.

this different scenario, RVEA-CAST still achieves a set of DEGs that are highly similar to those from the real clusters, as it can be observed by the circles of the Venn diagram (depicted in Fig. 8(a)). The accuracy and F1-score values obtained by RVEA-CAST (91.12% and 95.35%, respectively) also indicate that our approach outperforms the other methods. EMEP attains acceptable values probably as a result of the use of cluster ensembles (Fig. 8(c)), while MOGA results indicate that its DEGs are significantly different from those of the real clusters (Fig. 8(b)).

The Zeisel dataset is one of the most challenging in terms of different cell types with 48 of them. The Venn diagrams of RVEA-CAST and MOGA for this dataset are shown in Fig. 9. The absence of EMEP is due to the inability of this method to obtain results in a practical time period. In this dataset, the concordances of the DEGs from the real clusters with those from RVEA-CAST and MOGA are more similar, as it can be observed in the Venn diagrams. However, RVEA-CAST attains higher accuracy and F1-score values (92.48% and 96.09%) than MOGA (88.57% and 93.94%). Additionally, the number of genes in the intersection is higher (3970 DEGs out of 4009).

Regarding the Manno dataset, RVEA-CAST achieves higher accuracy and F1-score values (72.68% and 84.18%) than MOGA (69.28% and 81.85%) in the most complex instance in terms of cell types (Fig. 10). To further assess the biological relevance of the identified clusters, we conducted a deeper analysis of this dataset, which focuses on human ventral midbrain development, a topic of significant interest in Parkinson's disease research.

We performed Gene Ontology (GO) enrichment analyses of biological processes associated with the DEGs in the RVEA-CAST clusters. First, an enrichment analysis of all DEGs in the RVEA-CAST clusters was conducted to provide broader context. Among the top 10 enriched GO terms, we identified biological processes relevant to general development, such as *ribosome biogenesis* (GO:0042254) and *mitotic cell cycle phase transition* (GO:0044772), as well as processes specifically related to brain development, including *synapse organization* (GO:0050808), *regulation of neuron projection development* (GO:0010975), and *axoogenesis* (GO:0007409), among others (Fig. 11(a)).

RVEA-CAST effectively distinguished several groups of cells that significantly overlapped with known progenitor subpopulations in the real clusters. In some clusters, GO terms associated with proliferative processes, such as *DNA replication* (GO:0006260), were enriched. Additionally, the KEGG pathways *DNA replication* (hsa03030) and *cell cycle* (hsa04110) were significantly enriched. Among the DEGs, we identified key proliferation-related genes, including cyclins (*CCNB1*, *CCNB2*, *CCNA1*, *CCNE1*, *CCNC*), cyclin-dependent kinases (*CDK1*, *CDK2*, *CDK4*), and DNA polymerase genes (*POLA1*, *POLA2*, *POLD1*), highlighting the strong proliferative activity characteristic of progenitor populations.

In contrast, another progenitor cell cluster exhibited characteristics of a more advanced differentiation stage toward neural lineages, with enriched GO terms such as *modulation of chemical synaptic transmission* (GO:0050804), *regulation of synapse structure or activity* (GO:0050803), and *regulation of synapse organization* (GO:0050807), among others (Fig. 11(b)). Noticeably, DEGs in this cluster included neurotransmitter

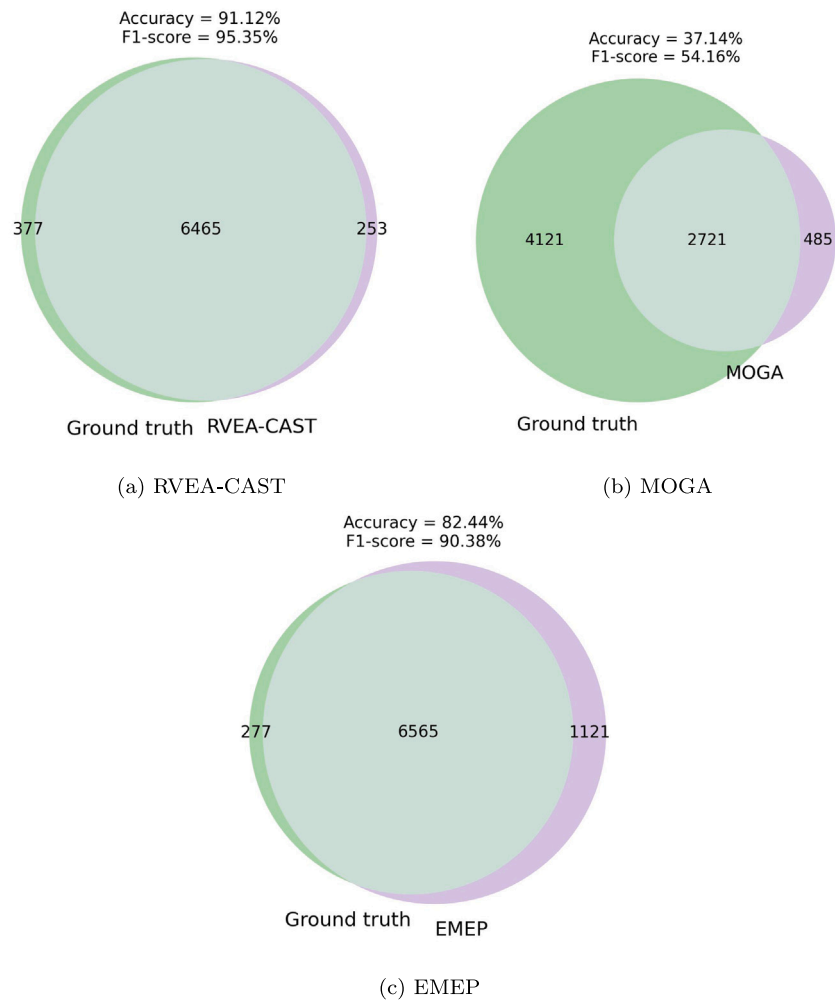


Fig. 8. Venn diagrams of the DEGs detected in the real clusters with those detected by RVEA-CAST (a), MOGA (b), and EMEP (c) in the Pollen dataset. Accuracy and F1-score of each method are also shown.

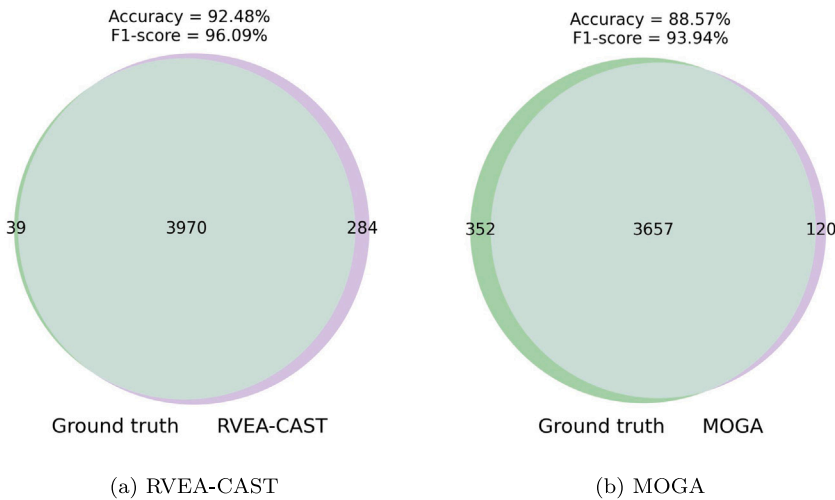


Fig. 9. Venn diagrams of the DEGs detected in the real clusters with those detected by RVEA-CAST (a) and MOGA (b) in the Zeisel dataset. Accuracy and F1-score of each method are also shown.

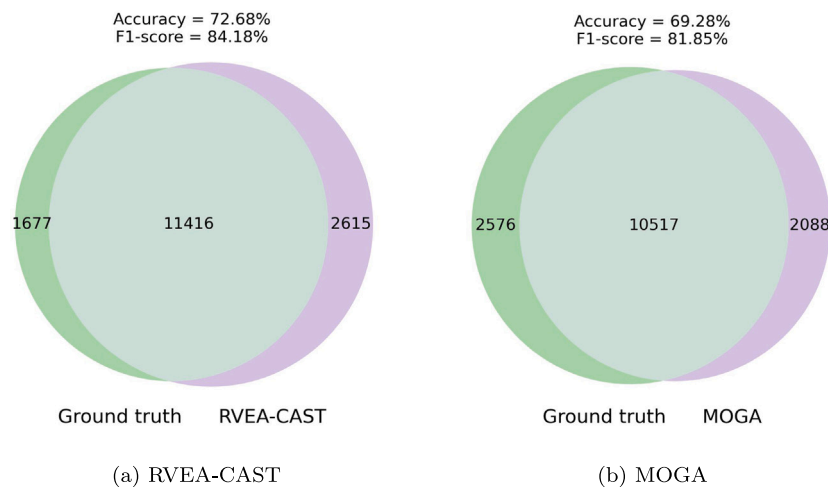


Fig. 10. Venn diagrams of the DEGs detected in the real clusters with those detected by RVEA-CAST (a) and MOGA (b) in the Manno dataset. Accuracy and F1-score of each method are also shown.

receptors (*CHRNA6*, *GABRG2*, *GRIA2*), neurotrophic factors (*BDNF*), and neurotransmitter release-related genes (*SNAP25*, *SYT1*). The enriched pathways included *dopaminergic synapse* (hsa04728), *Parkinson disease* (hsa05012), and *pathways of neurodegeneration — multiple diseases* (hsa05022), suggesting that these progenitor cells may be associated with dopaminergic neurons of the substantia nigra.

RVEA-CAST also identified a cluster with a significant overlap with radial glial cells. The enriched GO terms in this cluster included *gliogenesis* (GO:0042063), *glial cell differentiation* (GO:0010001), and *regulation of nervous system development* (GO:0051960). Interestingly, among the enriched KEGG pathways, we identified the *Notch signaling pathway* (hsa04330) (Fig. 11(c)), a well-established regulator of glial cell fate and differentiation.

Overall, this biological analysis highlights RVEA-CAST's ability to not only produce clusters that align more closely with real clusters than state-of-the-art methods but also to provide biologically meaningful clusters. These results indicate that RVEA-CAST successfully captures the underlying structure of scRNA-seq data, generating clusters of biological significance.

5.8. Summary of results and discussions

As shown in the previous sections, RVEA-CAST showcased superior performance compared to state-of-the-art methods in most of the 10 real scRNA-seq datasets analyzed. It outperformed existing approaches in eight out of ten datasets in terms of NMI and ARI. In particular, statistically significant improvements were observed in most comparisons, with increases of up to 66.7% for NMI and 261.5% for ARI. The importance of the intelligent mutation operators devised for RVEA-CAST was also evaluated, exhibiting statistically significant improvements over the version without these operators, particularly in the most complex clustering scenarios. Furthermore, a scalability study showed that the RVEA-CAST algorithm scales efficiently. Finally, the biological relevance of the clusters identified by RVEA-CAST was assessed, indicating a higher concordance with real clusters compared to state-of-the-art algorithms, as well as detecting enriched GO terms and KEGG pathways characteristic of the cell types under study.

6. Conclusions

Single-cell RNA-seq has revolutionized transcriptomic studies by enabling the measurement of gene expression at the single-cell level in biological samples. Although quantifying transcripts in individual cells presents diverse challenges, the invaluable insights gained into complex

and rare cell types, and cell to cell variability make it a worthwhile effort.

Clustering analysis plays a key role in identifying distinct cell populations in scRNA-seq data. This analysis aims to uncover cell subgroups within a biological sample. The creation of cell atlases, which aid in identifying new cell types and biomarkers of rare cell populations, stands out as one of the main applications of clustering analysis in scRNA-seq data. Due to the extended popularity of scRNA-seq, several clustering algorithms have been developed specifically for the data generated from this technology.

In this paper, a novel method for multi-objective scRNA-seq clustering has been proposed. This method, designated as RVEA-CAST, incorporates three problem-aware mutation operators designed to enhance each objective of the problem: clustering deviation, clustering compactness, and DBI. These mutation operators dynamically change the number of clusters of the solutions within a range specified in the initialization of the algorithm, overcoming one of the main challenges of clustering which is the selection of the optimal number of clusters. In addition, the multi-objective search engine of the proposal is based on the idea of handling different subspaces derived from the original objective space through the use of reference vectors.

The results obtained by RVEA-CAST have been compared with those from other evolutionary algorithms in the field using ten real scRNA-seq datasets, evaluating their performance by using the NMI and ARI metrics. Our comparisons revealed that RVEA-CAST outperforms the previously proposed algorithms in most datasets, showing statistically significant improvements of up to 66.7% for NMI and 261.5% for ARI. A biological perspective of the results has also been provided through differential gene expression analysis of the clusters predicted by the different methods and the real clusters. This analysis showed that our method consistently outperforms the others in terms of accuracy and F1-score taking as a reference the differentially expressed genes in the real clusters. These results were obtained from datasets with a wide range of cell types, showcasing the robustness of RVEA-CAST and its ability to obtain high-quality results in diverse biological scenarios.

As future work, we plan to devise alternative strategies to enhance the results for the most complex datasets. Preprocessing steps, such as batch effect correction, imputation of dropouts, and dimensionality reduction, can significantly impact clustering outcomes. Therefore, developing specialized workflows for datasets that exceed a certain cell number threshold could improve the results in these challenging scenarios. Furthermore, we propose to extend the application of advanced multi-objective optimization strategies to address other critical aspects of scRNA-seq analysis, particularly in generating marker panels for specific cell populations. The identification of robust marker panels would

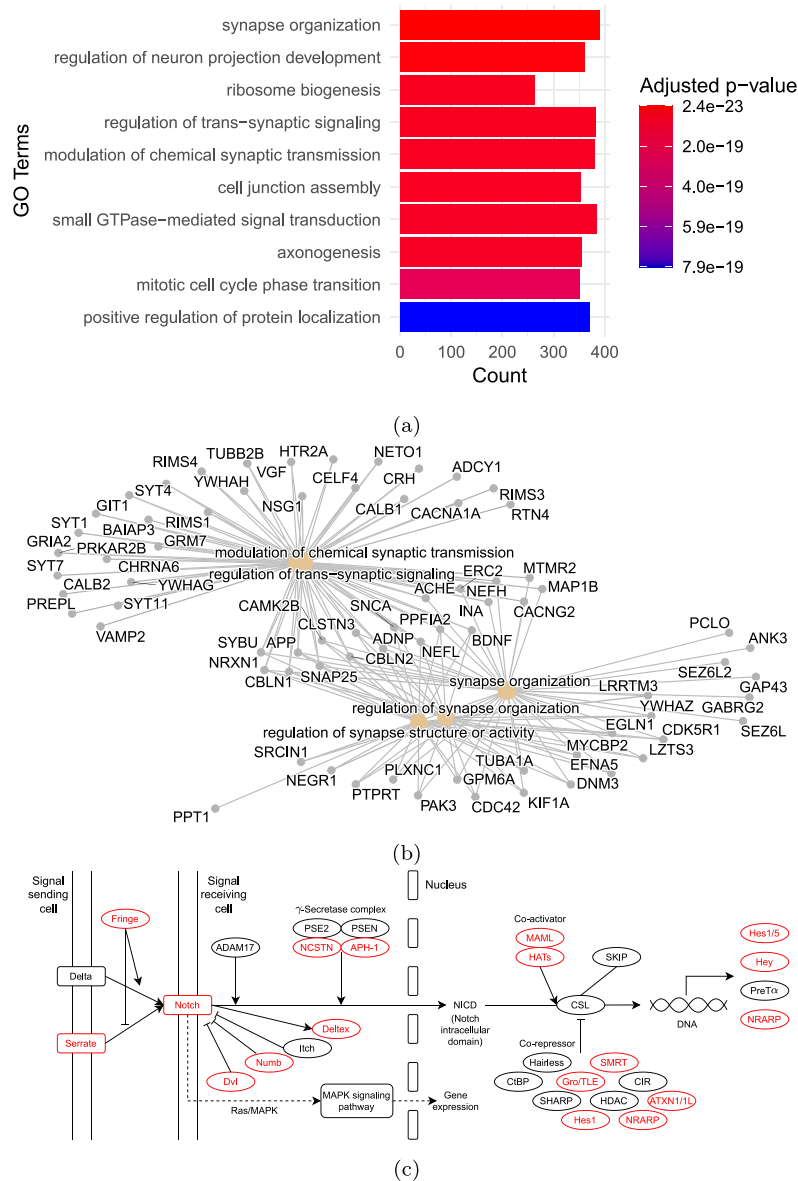


Fig. 11. Top 10 enriched GO terms in RVEA-CAST clusters (a). Network of genes and enriched GO terms in a progenitor cell population (b). Notch signaling pathway from KEGG, with DEGs highlighted in red (c).

facilitate the isolation of subpopulations identified during clustering, enabling more targeted downstream analyses which provide detailed insights into cellular heterogeneity and function.

CRediT authorship contribution statement

Fernando M. Rodríguez-Bejarano: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Miguel A. Vega-Rodríguez:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Sergio Santander-Jiménez:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Ethics statement

The authors confirm that an Ethics Statement is not required for this research work. All the data employed in this work was obtained from

public databases that comply with international Ethics guidelines and the results herein presented do not have implications from an Ethics perspective.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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